

The maximum possible value of g is achieved by selecting $k = 4$ and choosing the numbers $a_1 = g - 1$, $a_2 = 2g - 1$, $a_3 = 3g - 1$, and $a_4 = 4g - 1$. These numbers are distinct positive integers, and their sum is $10g - 4$, which must be strictly less than 3×10^6 . This inequality gives $g < 300,000.4$, so the largest integer g is 300,000.

For this choice of g , each $a_i \equiv -1 \pmod{g}$. The product $P = (g - 1)(2g - 1)(3g - 1)(4g - 1)$. For any i , P/a_i is the product of three terms each congruent to $-1 \pmod{g}$, so $P/a_i \equiv -1 \pmod{g}$. Thus, g divides $(P/a_i) + 1$ for all i . The sum $10 \times 300,000 - 4 = 2,999,996$ is less than 3×10^6 , satisfying the problem's conditions. No larger g is possible because the sum would exceed the limit.

\boxed{300000}